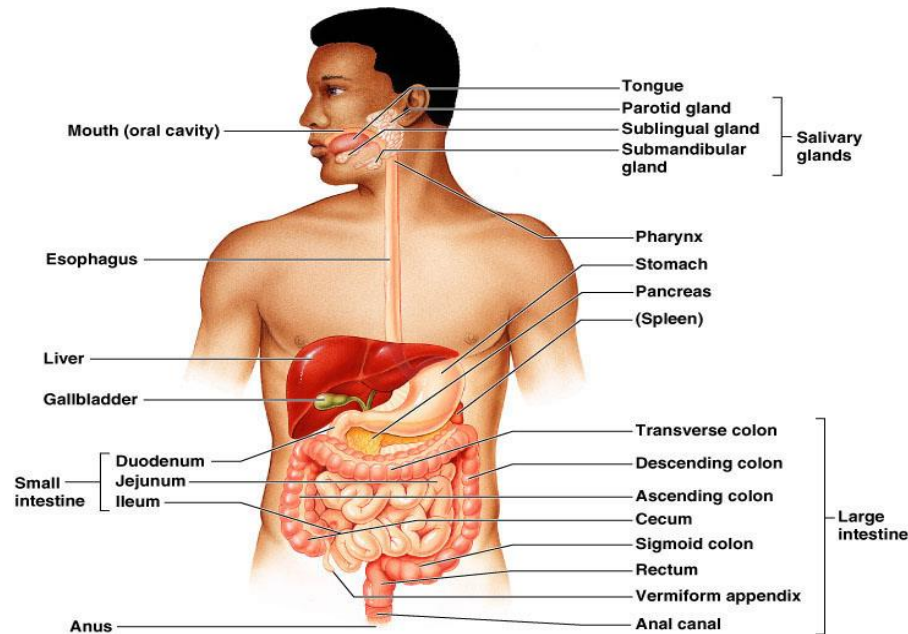


Physiology of the GIT



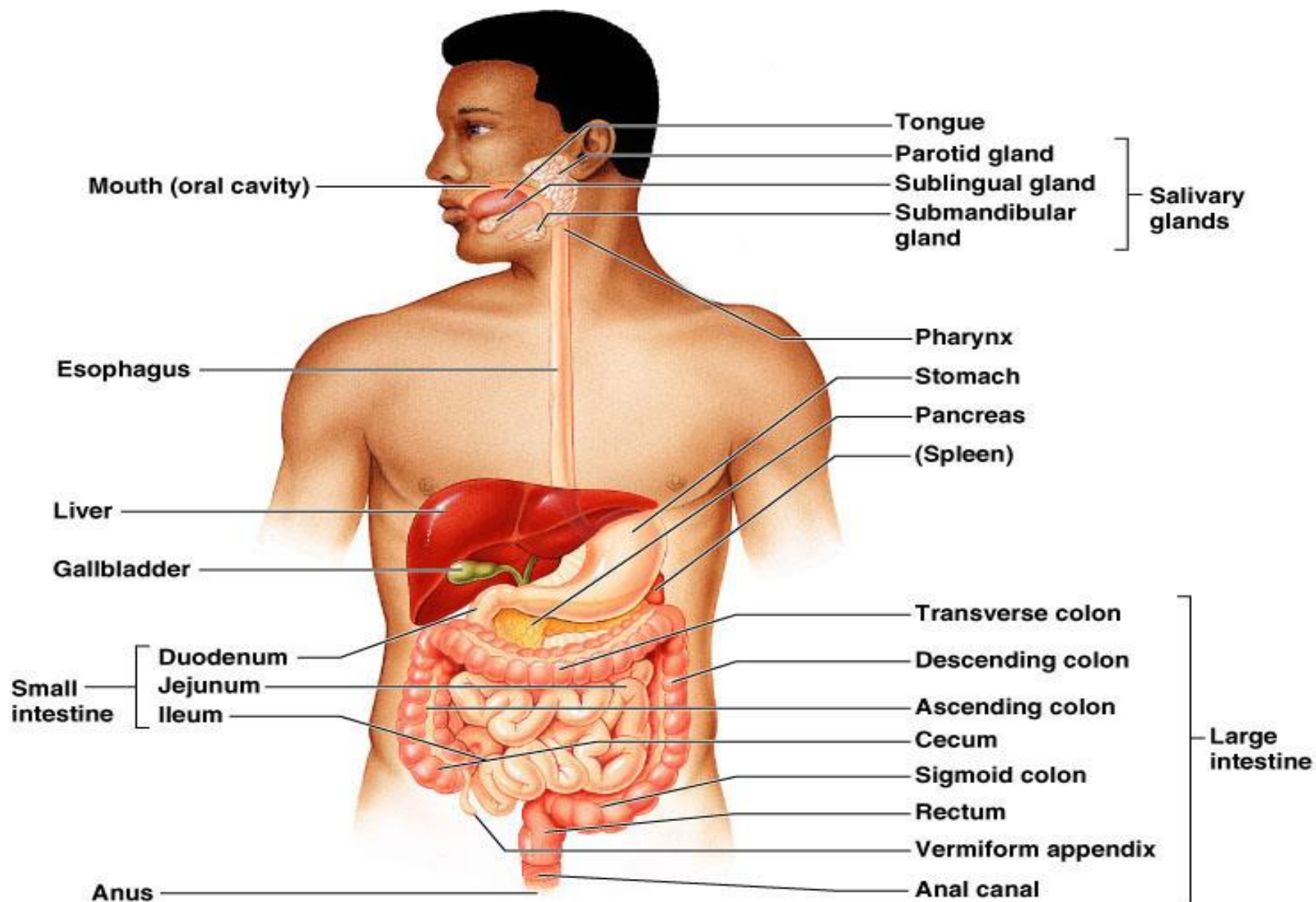
Introduction

Introduction

- **Functions:**
 - **Ingestion**/food intake → controlled by the feeding and satiety center in the Hypothalamus
 - **Mastication**/chewing → mechanical grinding of food with the aid of the teeth
 - **Swallowing**/deglutition → propulsion of food from the mouth to the stomach
 - **Chemical digestion** of food → digestion by enzymes
 - **Supplies energy nutrients**

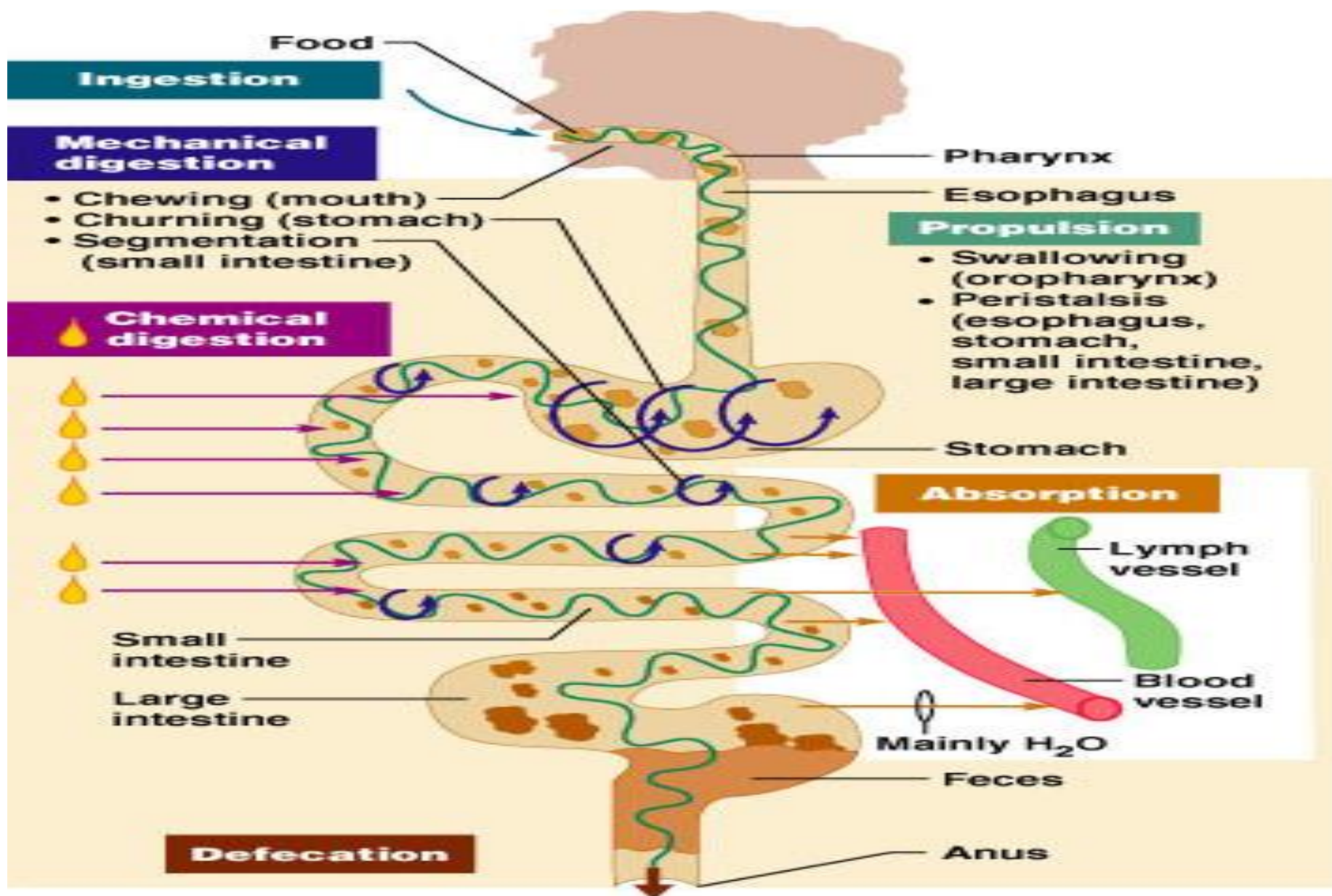
Introduction

- Functions:
 - Fluid balance → absorption + secretion + excretion
 - Microbial defense:
 - Mouth → Saliva contains lysozymes and Immunoglobulins (IgA)
 - Stomach → HCl has a bactericidal actions
 - Small intestine → Peyer's patches and macrophages
 - Secretion of enzymes, mucus, and hormones
 - Absorption of nutrients, water and electrolytes into the circulation
 - Defecation → excretion of fecal matter



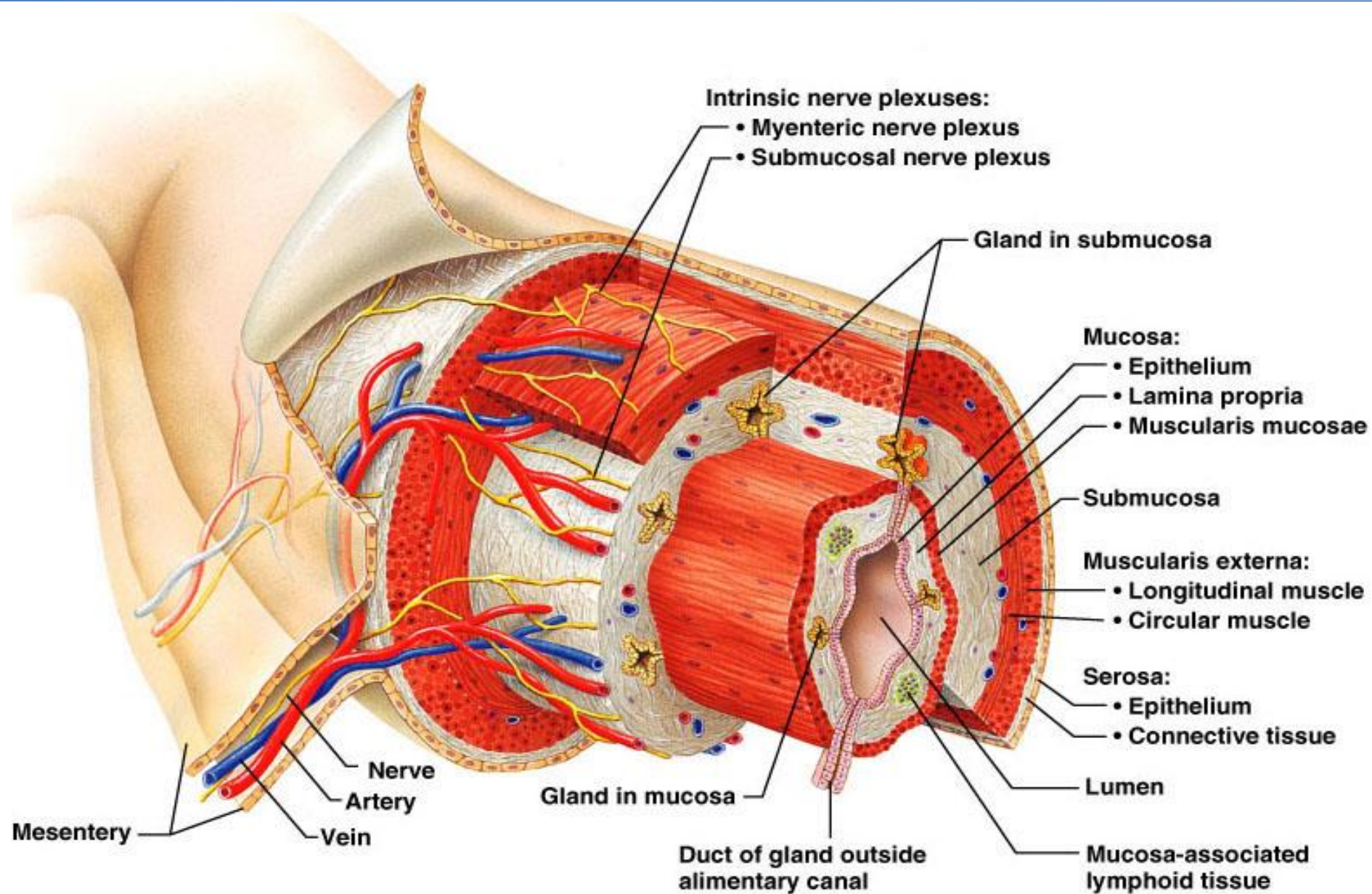
Introduction

- Digestive Process:
 - Mouth → Food ingestion controlled by the HT
 - Esophagus → Simple passage of food
 - Stomach → Temporary storage + Mixing+ Digestion + Secretion + Emptying
 - Small intestine → Digestion + Secretion + Absorption
 - Large Intestine → Storage + Absorption + Passage of faecal matter



Cytoarchitecture

- From esophagus to the anal canal the walls of the GIT have the same four layers
- From the lumen outward:
 1. Mucosa
 2. Submucosa
 3. Muscularis externa and
 4. Serosa
- Each layer has a predominant tissue type and a specific digestive function



Cytoarchitecture

1. Mucosa:

- Located in the interior directly facing the lumen of the GIT
- Functions:
 - Secretion of mucus
 - Absorption of end products of digestion
 - Protection against infectious disease
- Consists of 3 layers → a lining epithelium, lamina propria, and muscularis mucosae

Cytoarchitecture

- Epithelial Lining

- Two types of epithelial cells along the GIT

1. Non-keratinized stratified squamous epithelium:

- Mouth, esophagus and anal canal
- Function → protection

2. Simple columnar epithelium: through out the rest of the tract

- Function → absorption + secretion (mucus):
 - Protect digestive organs from digesting themselves
 - Ease food movement along the tract

Cytoarchitecture

- Epithelial Lining
 - Stomach & SI mucosa contain enzyme & hormone-secreting cells (entero-endocrine cells)
- Lamina Propria
 - Contains connective tissues, blood vessels and lymph vessels
 - Nourishes the epithelium and absorbs nutrients
 - Contains lymph nodes important in defense against bacteria
- Muscularis mucosae
 - Smooth muscle cells that produce local movements of mucosa

Cytoarchitecture

2. Submucosa:

- Consists of loose connective tissues, secretory glands, lymph nodes, blood vessels and larger nerves
- Submucosal plexus (Meissner's plexus):
 - Enteric nerve plexus
 - Controls **secretions** of the GIT

Cytoarchitecture

3. Muscularis externa:

- **Skeletal muscles:**
 - Mouth, pharynx, upper esophagus → voluntary swallowing
 - External anal sphincter → voluntary control of defecation
- **Smooth muscles** → the rest of the GI tract
 - Circular fibers → surrounds the sub-mucosal layer
 - Longitudinal fibers → surrounds the circular muscles

Cytoarchitecture

3. Muscularis externa:

- Contraction of circular muscle → decrease diameter of the lumen
- Contraction of longitudinal muscle → shortens the tube length
 - Help mix and propel luminal contents
- Myenteric (Auerbach's plexus):
 - Major nerve supply of muscularis externa
 - Controls GIT motility

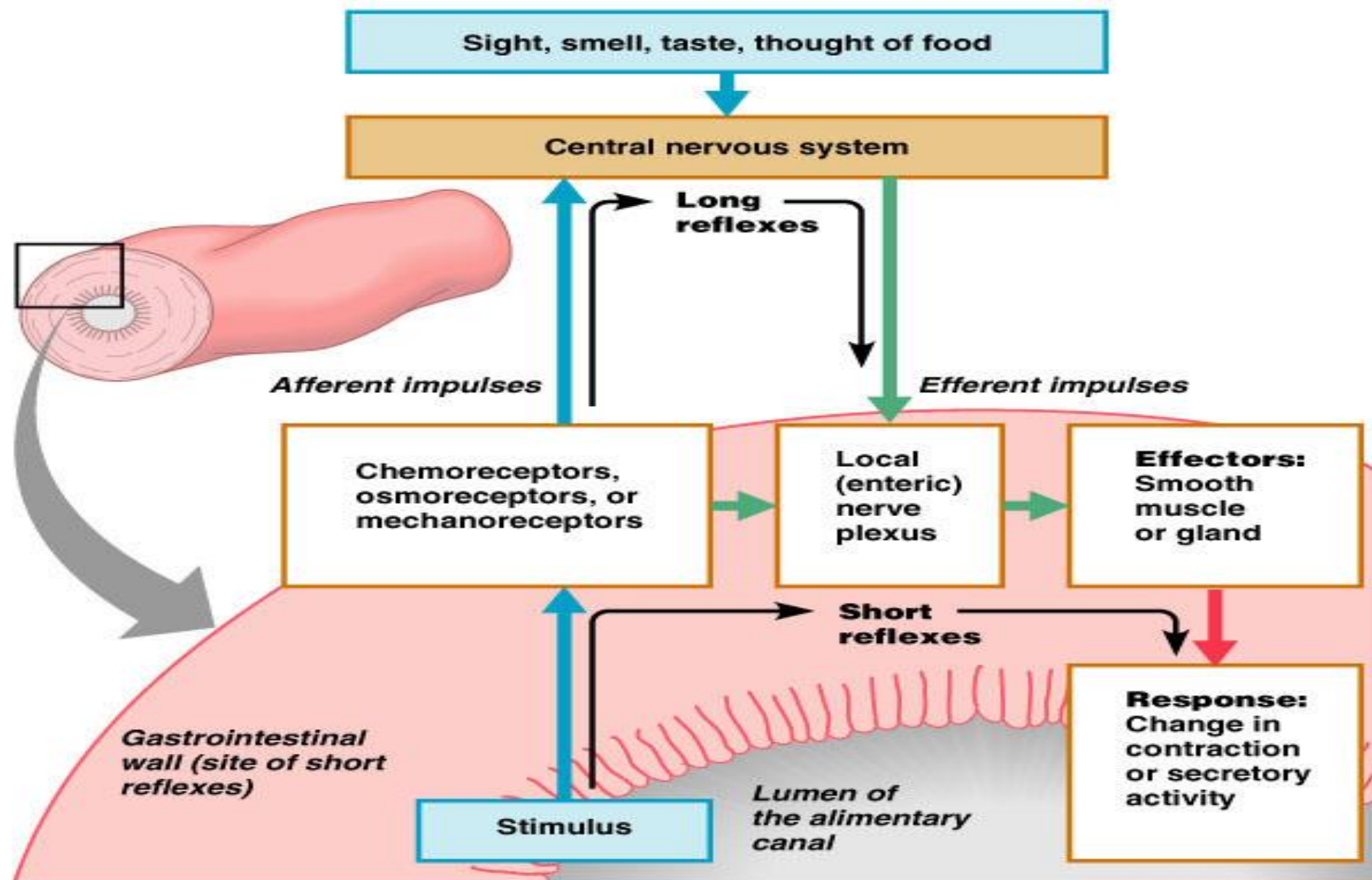
Cytoarchitecture

4. Serosa:

- The outer most layer of the GIT
- A serous membrane composed of connective tissue and epithelium
- Its attachment with the peritoneum helps in holding the gut in the abdominal cavity
- Function → protection

GI Regulation

- Receptors:
 - Mechanoreceptors → respond to distension, spastic contraction
 - Chemoreceptors → respond to osmolarity, pH, presence of fat and protein food and end products of digestion.
 - Thermoreceptors → respond to warm/cold food/drinks
 - Pain receptors → respond to tissue injury
 - They initiate **GI-reflexes**:
 - Activate or inhibit digestive gland secretions
 - Mix lumen contents and move them along the GIT



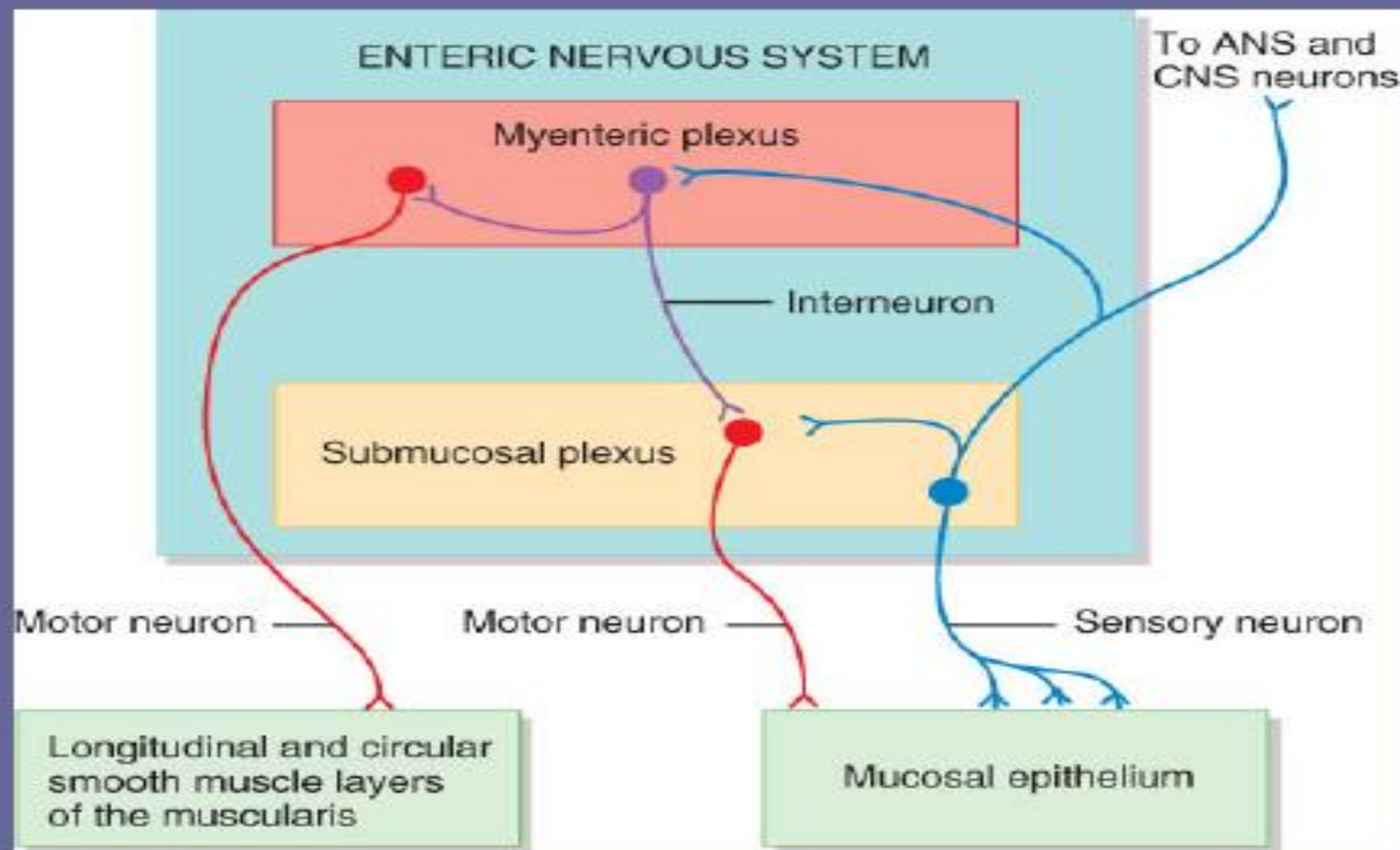
GI Regulation

1. Neural control:

1. ENS → Intrinsic control system
 - Submucosal (Meissner's plexus) → control GI secretion
 - Myenteric (Auerbach's plexus) → control GI motility
2. ANS → Extrinsic control system
 - SNS → Inhibit GI function
 - PNS → Increase GI activity

2. Hormonal control → GIT hormones

Enteric Nervous System

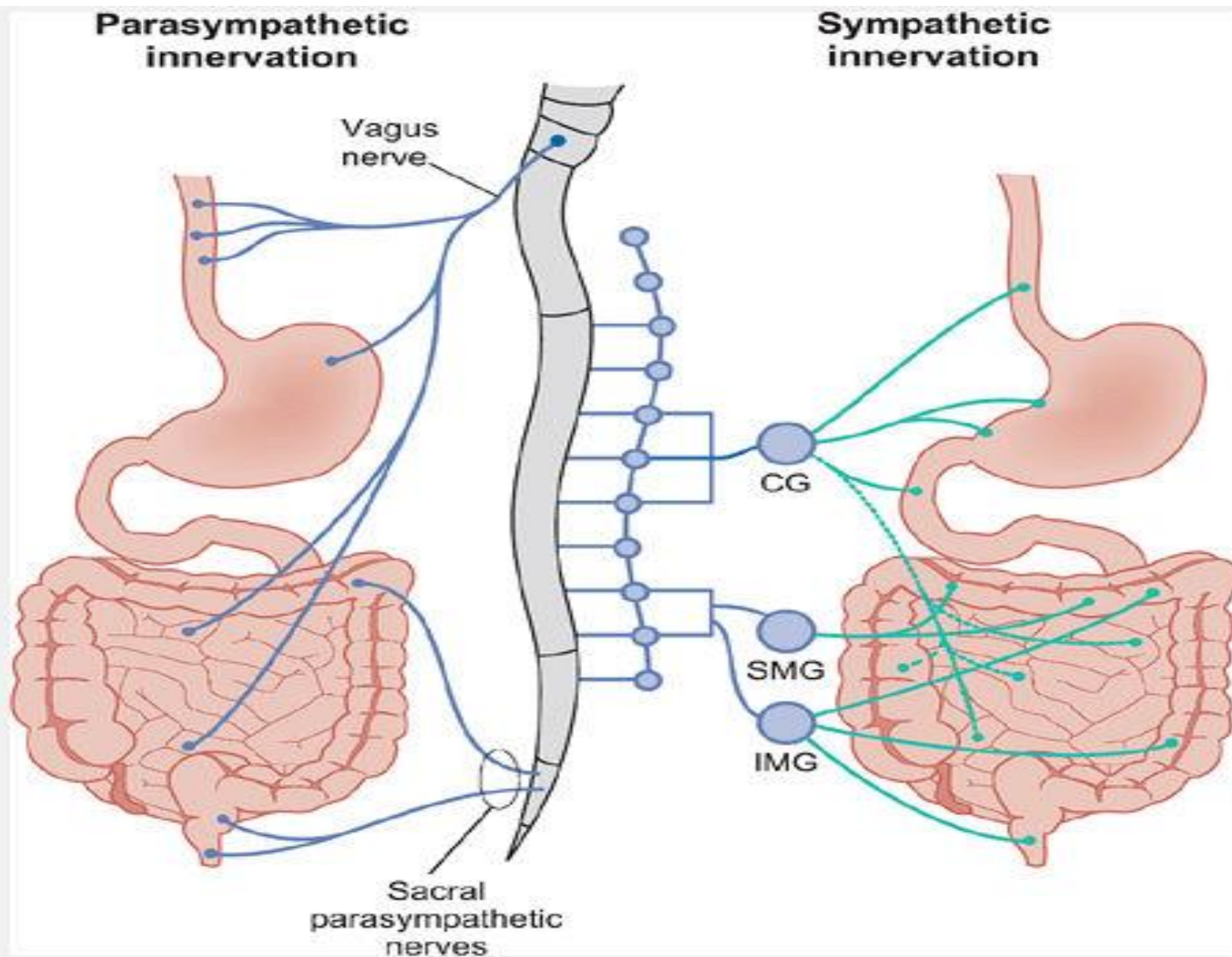


GI Regulation

- Parasympathetic innervation
 - Cranial (Vagus) → esophagus, stomach, pancreas, and first half of the large intestine (*little innervation to the SI*)
 - Sacral (S2-4) → sigmoidal, rectal and anal regions
 - Control defecation reflexes
 - The postganglionic neurons are located in the myenteric and submucosal plexuses
 - Stimulation of PNS causes a general increase in activity of the entire enteric NS → enhance GI function

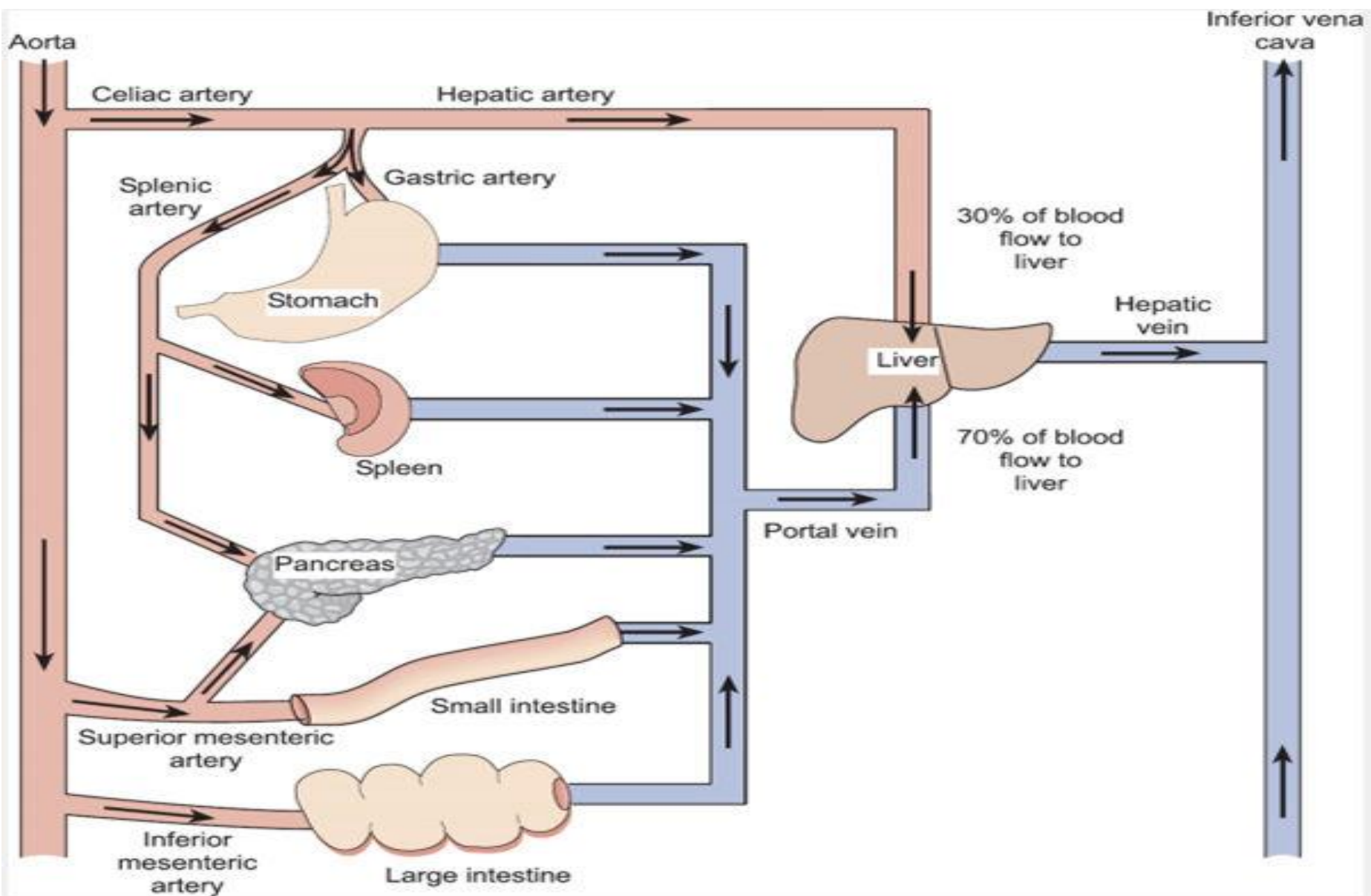
GI Regulation

- Sympathetic innervation
 - From T5 - L2
 - Preganglionic fibers enter the sympathetic chains and pass through the outlying ganglia → the celiac ganglion and mesenteric ganglion
 - Postganglionic fibers terminate in the enteric nerve plexuses
 - Innervate most portion of the GIT
 - In general, stimulation of SNS inhibit activities of the GIT
 - Strong stimulation of SNS can totally block GIT motility



Splanchnic Circulation

- **Supplies the** GIT + Spleen + Pancreas + Liver
 - Splanchnic BF = 1000 ml/min
- **Entero-hepatic circulation**
 - In the liver, blood passes through millions of liver sinusoids and finally leave the liver by way of the hepatic veins that empty into the inferior vena cava
 - **Reticuloendothelial** cells in the liver remove bacteria and other particles that entered the blood from the GIT thus preventing the direct access of potentially harmful agents into the systemic circulation
- Blood flow to the GIT increases by 100% during meal time



Splanchnic Circulation

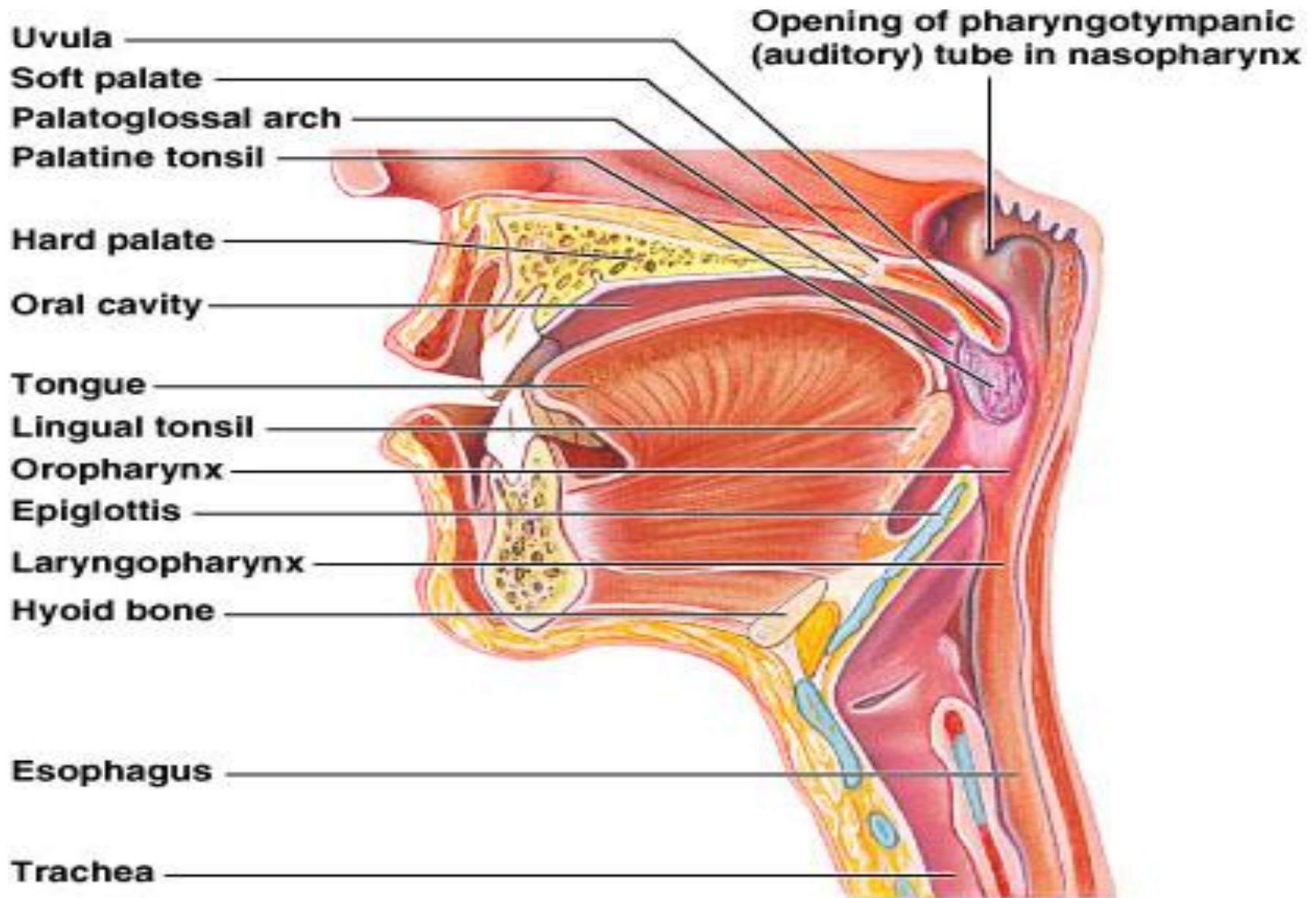
- Factors that control blood flow to the GUT:
 1. **Metabolic** → Increase in metabolism during or after a meal increases the release of vasoactive substances that increases blood flow (lactic acid, CO₂, H⁺, adenosine...)
 2. **Neural** → PNS - increases blood flow
→ SNS - decreases blood flow
 3. **Humoral**:
 - Serotonin, histamine, EP/NE → inhibit blood flow
 - Bradykinin, substance-P, adenosine → enhance blood flow

Physiology of the GIT

Oral Cavity

Mouth

- Oral/buccal cavity:
 - Bounded by lips, cheeks, palate and tongue
 - Lined with stratified squamous epithelium → withstand abrasions
 - The gums, hard palate and dorsum of the tongue are slightly keratinized
- Palate:
 - Roof of the mouth
 - Separate the mouth from the nasal cavity
 - Consists two portions → hard + soft palate



Mouth

- **Hard palate**
 - Bony structure at the roof of the mouth → assists chewing
- **Soft palate**
 - Mobile fold at the rear portion of the roof of the mouth
 - Formed mostly of skeletal muscle
 - Extend from the hard palate at the front + taper to the hanging uvula at the rear
 - It elevates to close off the nasal passages during swallowing
- **Uvula** → a small fleshy "V"-shaped extension of the soft palate that hangs above the tongue at the entrance to the throat

Mouth

- Tongue:
 - Occupies the floor of the mouth
 - Functions:
 - Repositioning food during chewing
 - Mixing food with saliva and forming the bolus
 - Initiation of swallowing, and speech
 - Contains taste buds (gustation)

Salivary Glands

1. Parotid gland (25%)

- Lies inferior + anterior to the ears b/n the skin and the masseter muscle
- Secrete almost entirely serous fluid rich in **Ptyalin**

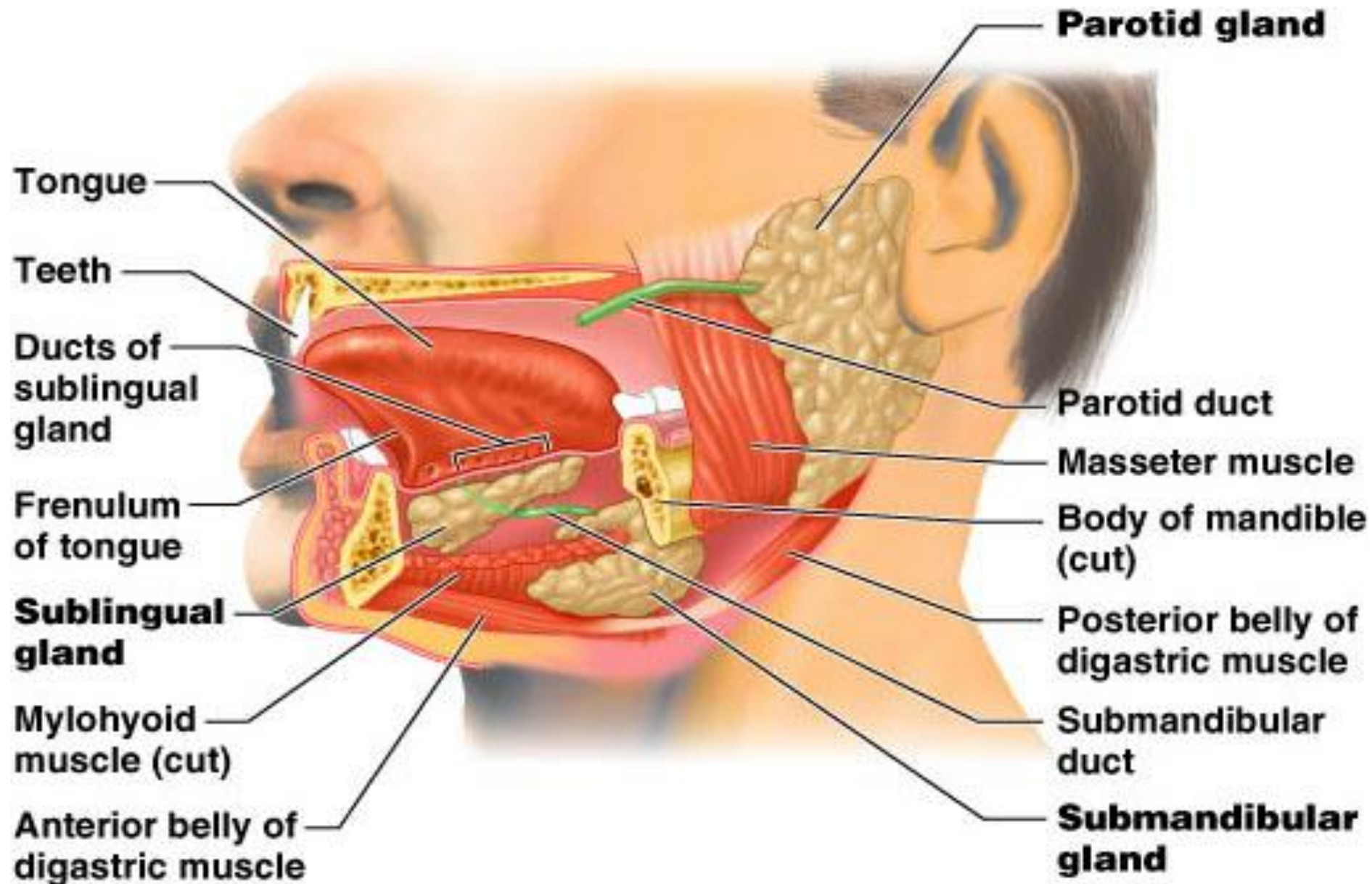
2. Submandibular gland (70%)

- Located at posterior floor of the mouth beneath the base of the tongue

3. Sublingual gland (5%)

- Located superior to the submandibular gland
- Produce both serous and mucus fluid like submandibular gland

4. Buccal gland → secrete only mucus



Salivary Secretion

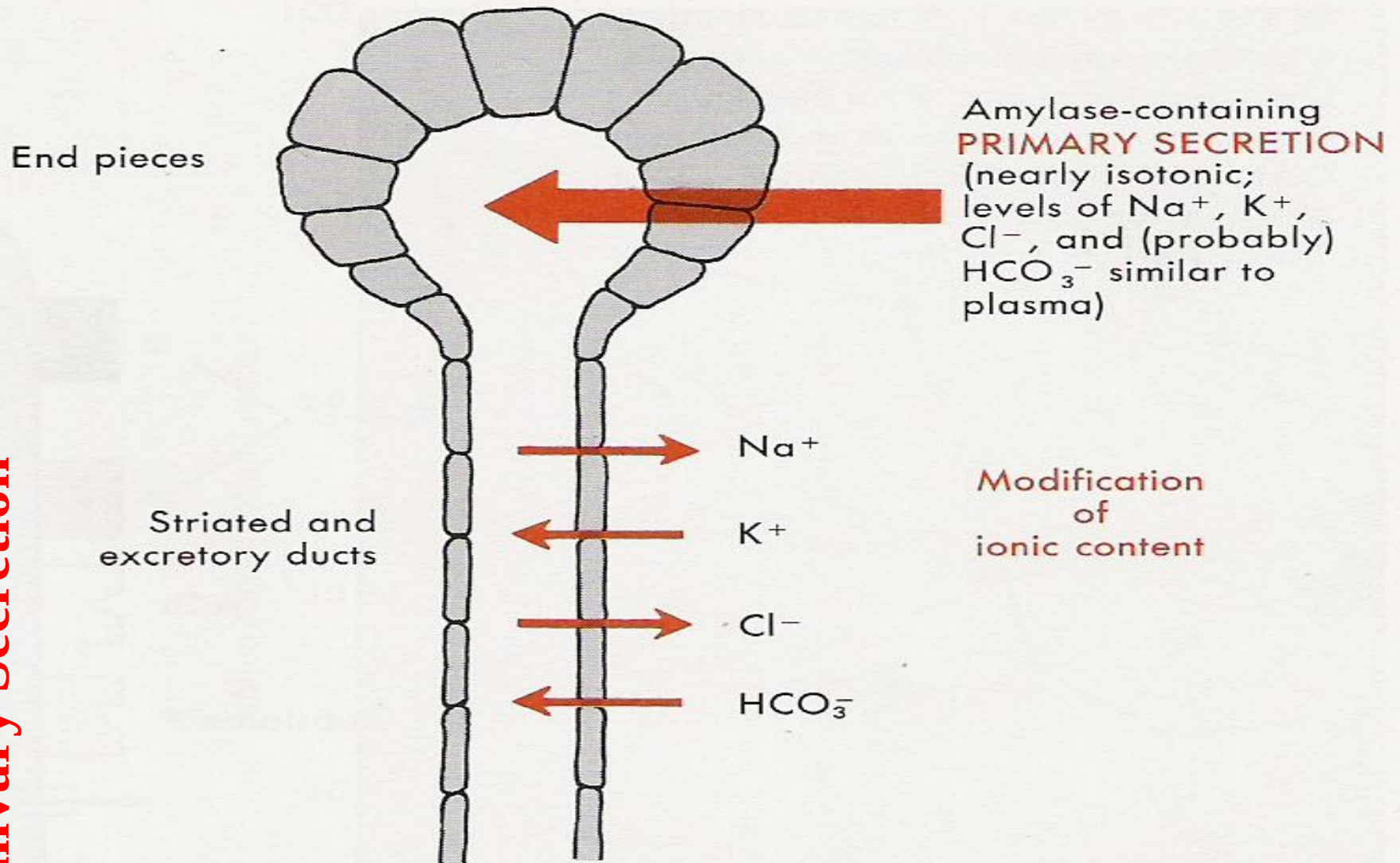
- Characteristics of saliva
 - Daily secretion $\approx$ 1000ml
 - Contain two types of protein secretions:
 - a. Serous secretion $\rightarrow$ contains ptyalin (α -amylase) $\rightarrow$ digest starches
 - b. Mucus secretion $\rightarrow$ contains mucin $\rightarrow$ lubrication + surface protection
 - pH $\rightarrow$ 6-7 $\rightarrow$ favorable for the action of ptyalin
 - Hypotonic ($\approx$ 97-99.5% water)
 - Electrolytes $\rightarrow$ Na^+ , K^+ , Cl^- , HCO_3^- ...

Salivary Secretion

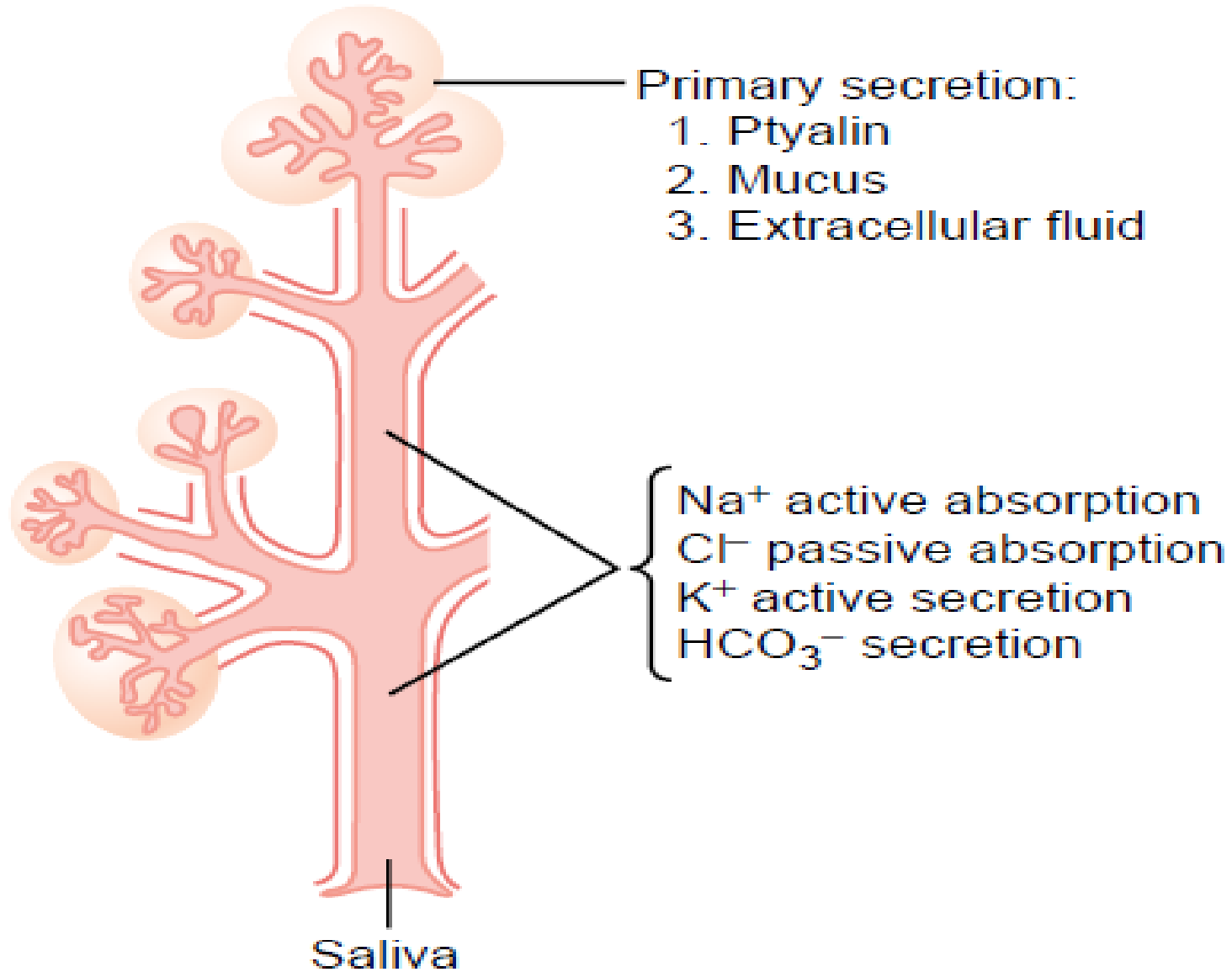
- Mechanism of saliva secretion
 - Saliva contains large quantities of $K^+ + HCO_3^-$ and smaller amount of $Na^+ + Cl^-$ than plasma
 - 1. Acini produce *primary isotonic fluid* (electrolytes, enzymes, proteins...) and secrete it into the ducts
 - 2. Salivary ductal cells modify ionic content of the 1° secretion:
 - Actively reabsorb Na^+ and secrete K^+
 - Na^+ reabsorption $>$ K^+ secretion $\rightarrow$ creates electrical negativity of $\approx -70\text{mv}$ in ductal cells $\rightarrow$ this drives passive reabsorption of Cl^-

Salivary Secretion

- Mechanism of saliva secretion
 3. HCO_3^- is passively secreted into the lumen in exchange for Cl^- (partly by active secretion)
- Ductal cells are relatively impermeable to water
 - Hence, H_2O does not follow the osmotic gradient of NaCl
- These causes *secondary hypotonic secretion*



Salivary Secretion

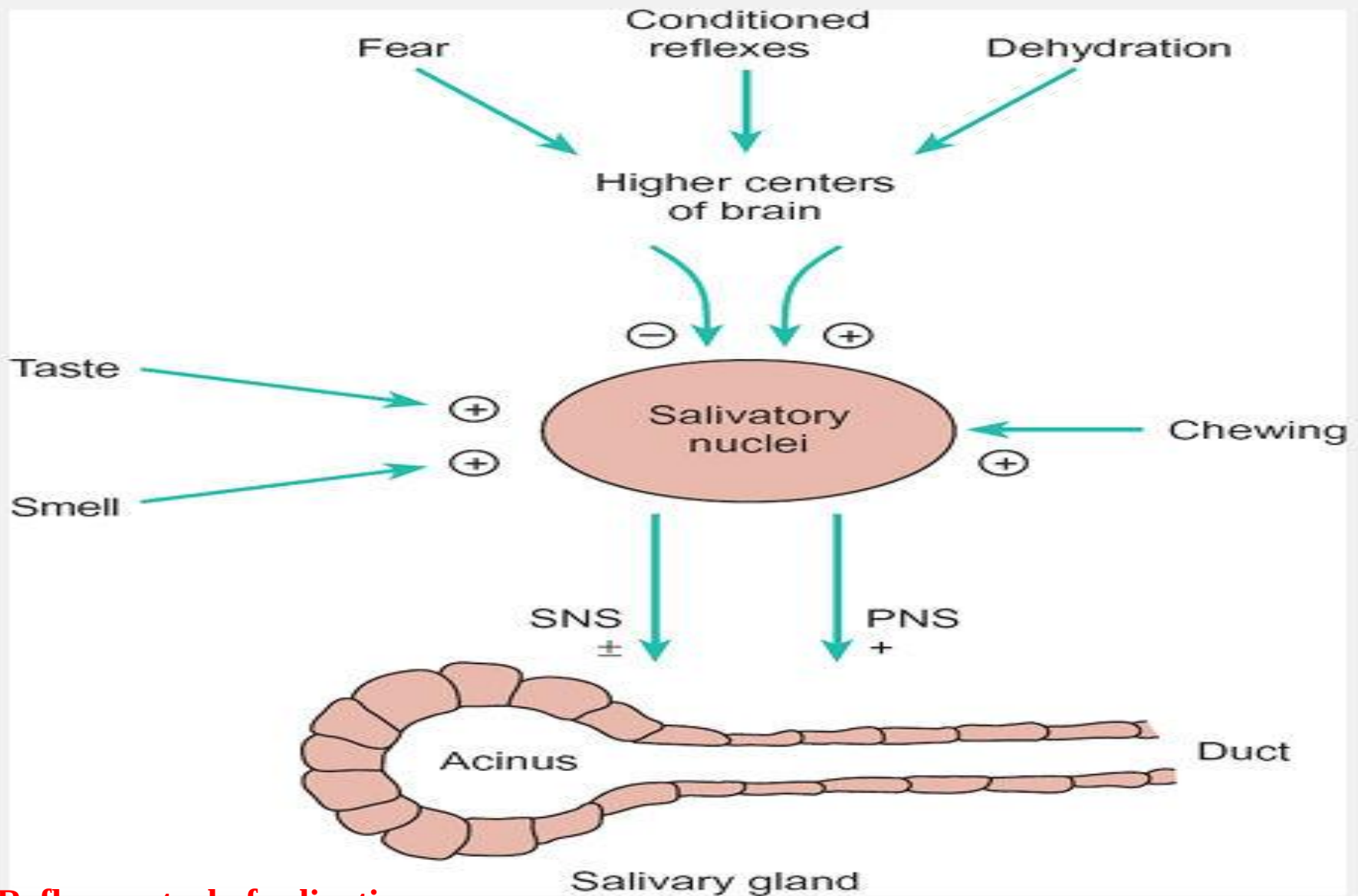


Salivary Secretion

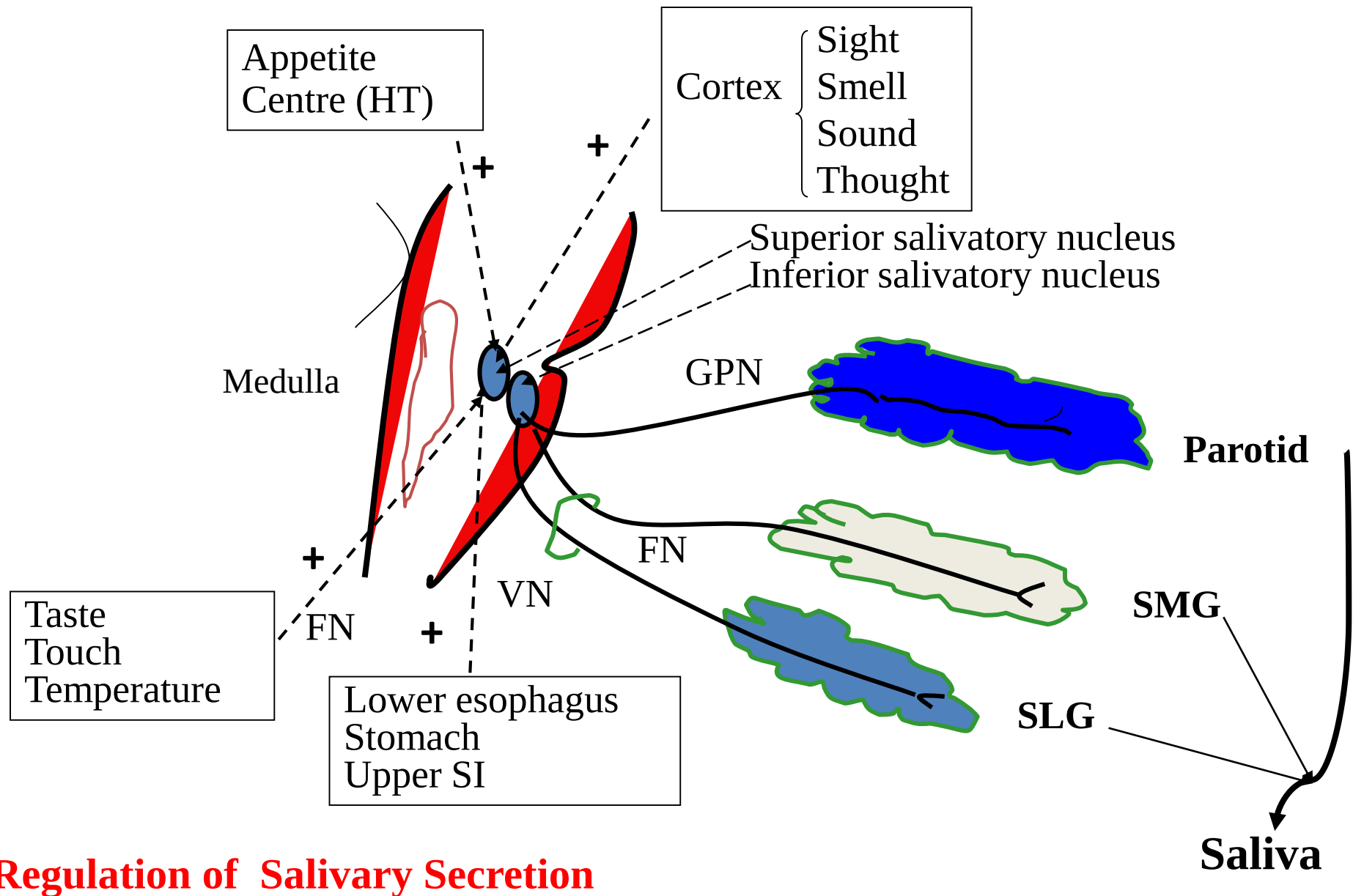
- Neural Regulation:
 - Salivary glands are mainly innervated by parasympathetic fibers from the superior and inferior salivatory nuclei in the brain stem
- Salivatory nuclei:
 - Located at the juncture of medulla and pons
 - Excited by taste and tactile stimuli from the mouth and pharynx (+ *smell, thought of food*)
- PNS stimulation causes copious secretion of watery saliva
- SNS stimulation causes small secretion of viscous saliva

Salivary Secretion

- Phases of salivary secretion:
 1. Cephalic (brain) phase:
 - Triggered by the thought, smell, or sight of food
 2. Oral phase:
 - Triggered by food stimulating touch and taste receptors in the mouth.
 3. Gastric phase:
 - Triggered by substances, which stimulate the GI mucosa

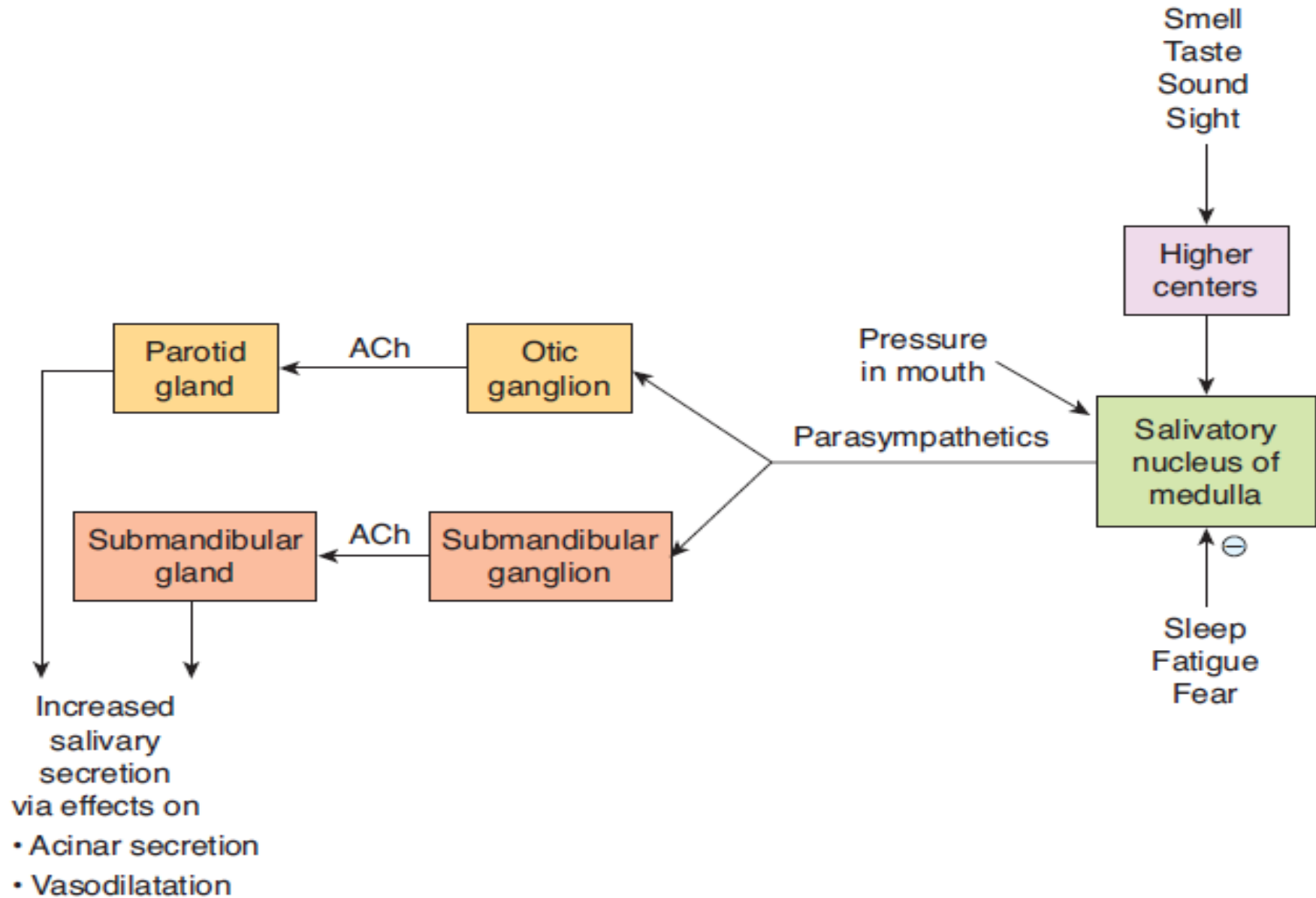


Reflex control of salivation



Regulation of Salivary Secretion

Regulation of Salivary Secretion

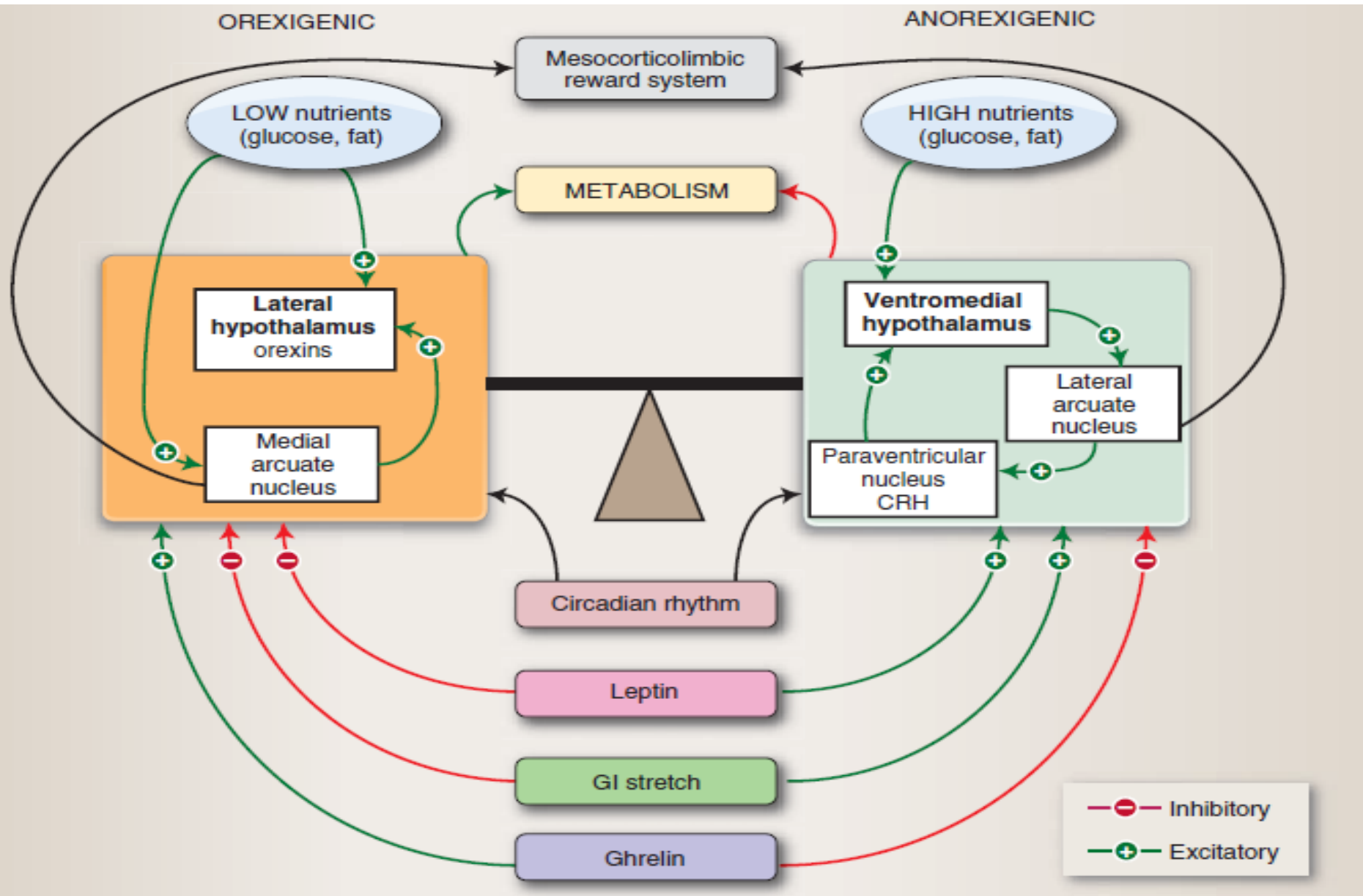


Salivary Secretion

- Functions of saliva
 - Moisten food → facilitate chewing and swallowing
 - Speech articulation → moistening allows lubrication and maneuverability of the tongue
 - Ptyalin → starch digestion
 - Thiocyanic ion (SCN^-) + Lysozymes → antibacterial actions
 - IgA → antibacterial actions
 - HCO_3^- → maintain a neutral pH favorable for ptyalin activity

Mouth

- Regulation of food intake (HT):
 - Lateral HT → Hunger/Feeding center
 - Stimulation causes craving for food and the desire for a particular type of food
 - Ventromedial HT → Satiety center → Inhibit the appetite center
- **NB**
 - They have reciprocal connection with the amygdala which play a role in emotional affective responses to food



Mouth

- Digestive processes:
 - Mastication → Mechanical digestion
 - Mastication muscles are controlled mainly by the trigeminal nerve
 - Salivary amylase → begins chemical digestion of starch
 - Bolus propulsion is initiated by swallowing

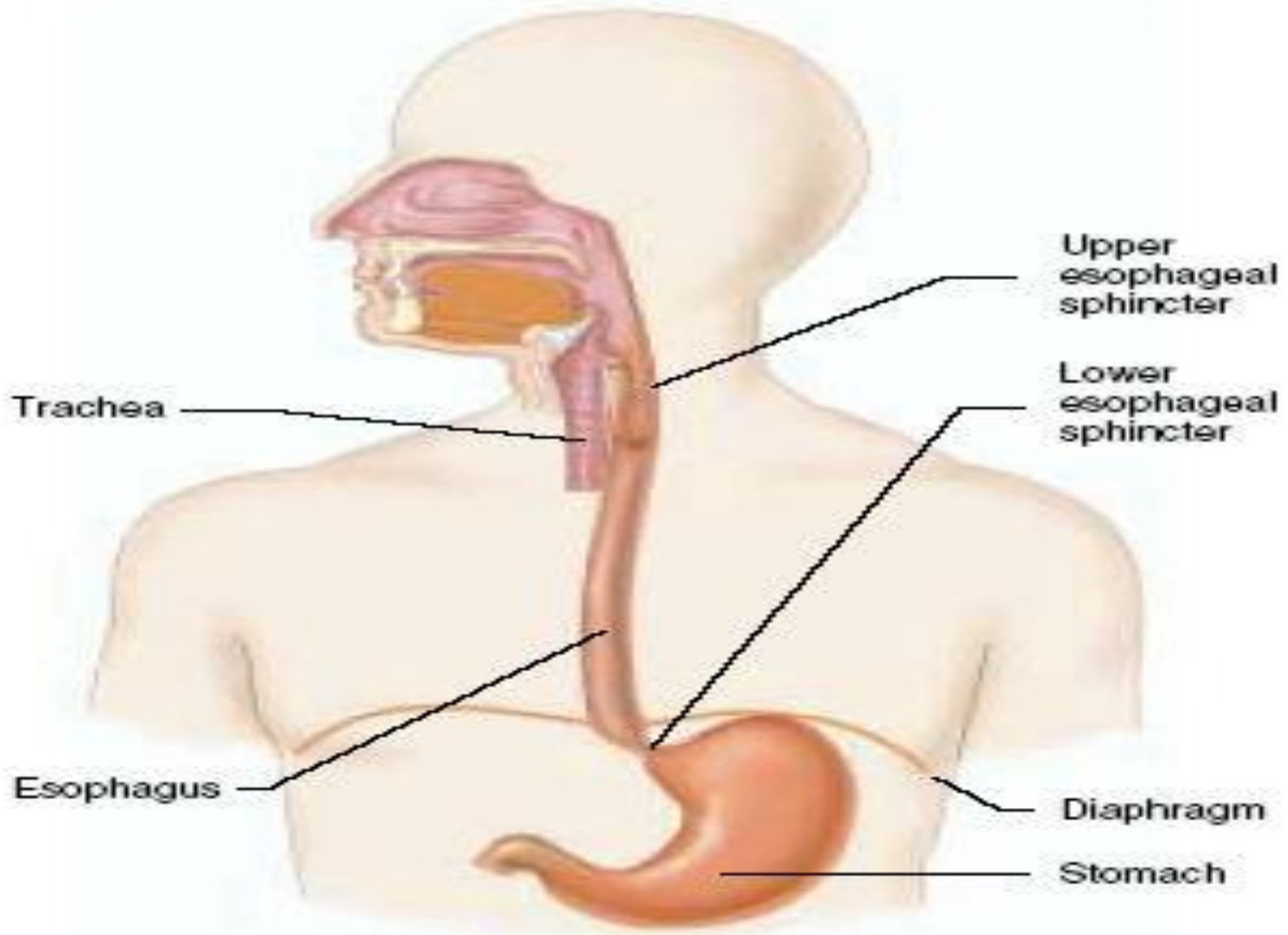
Pharynx

- Muscular tube located in the neck, lined with mucous membrane.
- Connects the nose and mouth with the trachea and esophagus:
 - Serves as a passageway for both air and food.
- Contains the **tonsils**, ends at the epiglottis.
- **Epiglottis** → a flap of cartilage, allows food to enter the esophagus & prevents it from entering the trachea.
- **Eustachian tubes (2)** → connect the pharynx with the middle ear
 - Help equalize air pressure that build up on the eardrums

NB: The pharynx is subject to infections entering through nose/mouth.

Esophagus

- A muscular tube passing behind the trachea and heart, and penetrates the diaphragm before reaching the stomach at the cardiac orifice.
 - Esophageal sphincter (circular muscle) → separate the esophagus and the stomach.
 - **Muscularis** changes from skeletal (superior third) to smooth muscle (inferior two-third)
- The pharynx and esophagus serves as conduits to pass food from the mouth to the stomach



Esophagus

- Esophageal mucosa:
 - Contain mucus glands → entirely produce mucous secretion as the bolus moves through the esophagus
 - Provide lubrication for swallowing
 - Prevents mucosal excoriation by newly entering food
 - Protect esophageal wall from digestion by reflux gastric acid

Deglutition

- The process of moving bolus from the mouth down to the stomach
- Has three stages:
 1. Voluntary stage
 2. Pharyngeal stage (involuntary)
 3. Esophageal stage (involuntary)

1. Voluntary stage:

- Oral stage of swallowing
- Food is chewed and mixed with saliva
- The tongue rolls and pushes the chyme into the pharynx

Deglutition

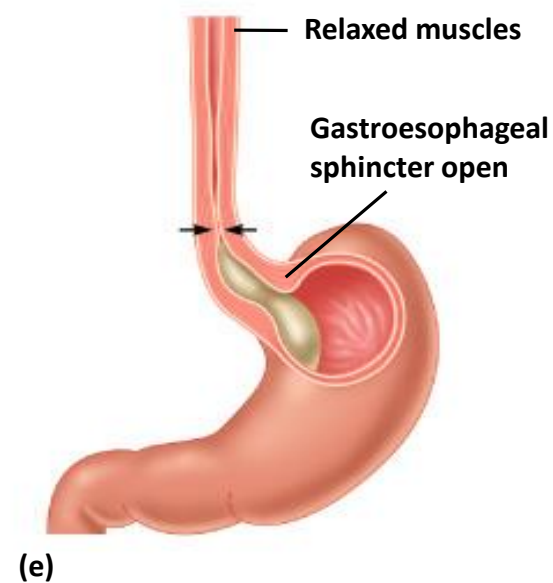
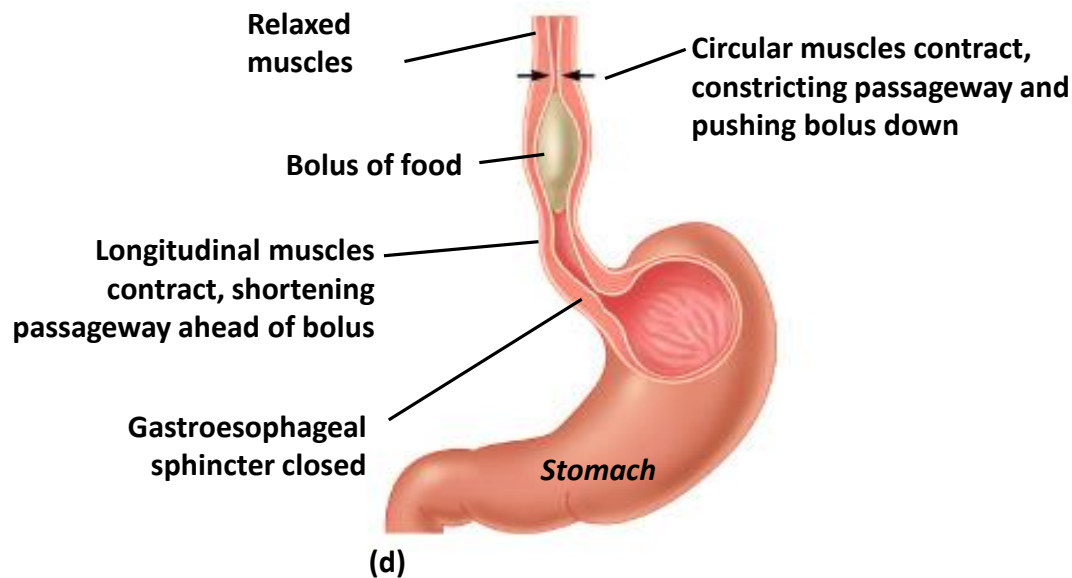
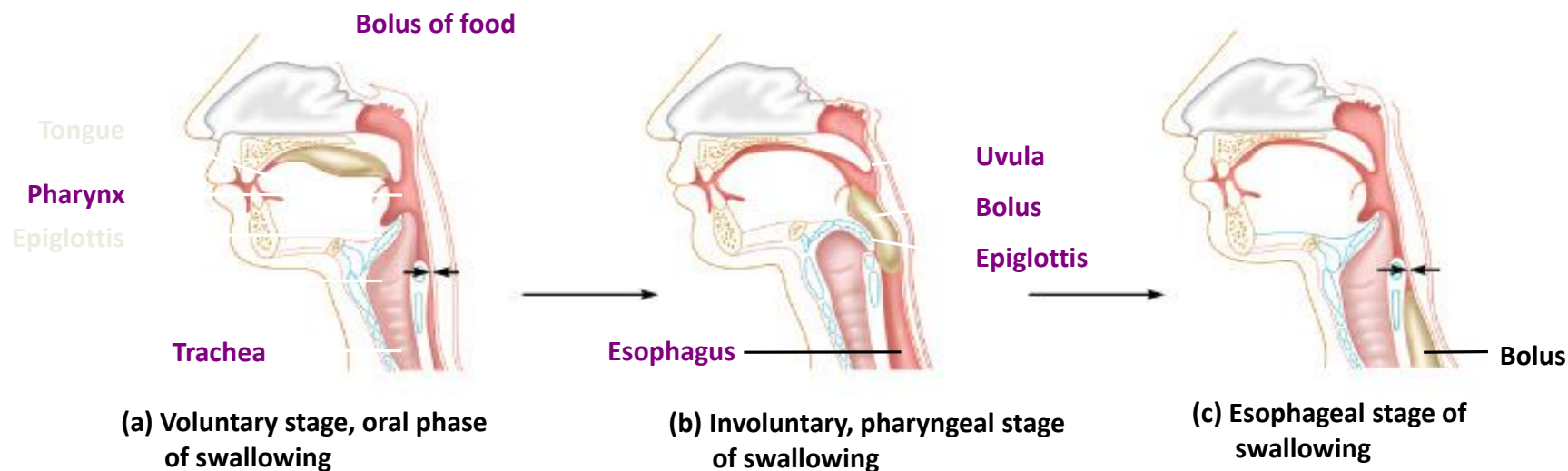
2. Pharyngeal stage:

- Mechanoreceptors in the pharynx are stimulated by the bolus of food → signals to the swallowing center (medulla) → pharynx:
 - Nasal opening is closed by the soft palate
 - Tracheal opening (glottis) is closed by upward movement of the tracheal muscles → *temporary inhibition of respiration*
- The bolus is then pushed into the esophagus by peristaltic waves started in the pharynx

Deglutition

3. Esophageal stage:

- The bolus is continuously pushed by peristalsis into the stomach by passing through the opening of the lower esophageal sphincter
 - *1° peristalsis* → continuation of peristalsis initiated in the pharynx
 - *2° peristalsis* → additional peristaltic wave initiated by the bolus of food distending the esophageal wall
- **Gastro-esophageal sphincter (LES):**
 - Thickened circular smooth muscle at the junction of esophagus & stomach → prevent reflux of gastric contents into the esophagus



Deglutition

- **Gastro-esophageal reflux:**
 - Entry of gastric contents into the lower part of the esophagus
 - Caused by incompetence of the LES
 - Leads to ulceration of the lower esophageal mucosa
- **Achalasia:**
 - Inability to swallow food due to failure of LES to relax
 - Caused by increased tone of LES due to high sensitivity to gastrin, weak esophageal peristalsis....